

ADHITHYAN S

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering undergraduate specializing in full-stack development and applied GenAI/RAG systems. Built a Python automation tool at Hitachi Energy cutting manual processing time by **25%**, and engineered end-to-end RAG pipelines for document retrieval and voice automation. Microsoft Azure and Salesforce certified professional.

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript (ES6+), C, C++

Web & Databases: React.js, Next.js, Node.js, FastAPI, HTML5/CSS3, PostgreSQL, MongoDB

AI Technologies: GenAI, RAG, LangChain, Vector Search (Qdrant, ChromaDB), REST APIs, LLMs

Dev Tools & Cloud: VS Code, Docker, Postman API, Git/GitHub, Microsoft Azure, Salesforce

CS Fundamentals: DSA, OOP, OS, DBMS, System Design (Basics)

INTERNSHIP

Hitachi Energy Technology Services Private Limited – Semiconductor Team

Chennai, India

Student Intern

June 2025 – July 2025

- Developed a **Python automation script** converting raw, unformatted Excel data into a structured template for internal engineering tools, cutting manual processing time by **25%**.
- Progressively extended the script from basic file/data entry to **column comparison, test-name insertion, and multi-file batch processing**, debugging data alignment and validation issues across raw datasets.
- Collaborated with mentor and team through daily Agile scrums, gaining exposure to R&D, quality testing, and production workflows within the Semiconductor division.

PROJECTS

JARVIS Multimodal Voice Assistant (GitHub) | *Python, Groq API, SQLite, Edge-TTS*

- Engineered a low-latency, event-driven voice pipeline integrating **SpeechRecognition**, the **Groq LLM API**, and **Edge-TTS**, achieving a **2.5s median end-to-end response time** across a 10-command benchmark suite.
- Replaced rigid regex command-matching with **dynamic, prompt-driven OS dispatching**, achieving **80% first-pass action-dispatch accuracy** across Win32 APIs, PowerShell commands, and native applications.
- Built persistent conversational memory via **SQLite** and multimodal screen understanding via **PyAutoGUI**, with multi-threaded global hotkeys for instant speech interruption.

Lexicon Ai (GitHub) | *Next.js, Supabase (PostgreSQL + pgvector), Gemini API*

- Engineered a full-stack **Retrieval-Augmented Generation (RAG)** platform combining a **Manifest V3 browser extension** with a **Next.js** application to capture and semantically index web content on demand.
- Implemented a **hybrid search layer** on Supabase (Postgres + pgvector) fusing full-text and vector similarity retrieval via **Reciprocal Rank Fusion**, achieving **90% top-1 retrieval relevance** across tested queries.

AI Research Paper Explainer (GitHub) | *FastAPI, Streamlit, Gemini API, ChromaDB*

- Engineered a **RAG pipeline** using FastAPI and Google Gemini for page-cited semantic Q&A over dense academic PDFs, achieving **68.8% exact page-citation accuracy** and a **0% hallucination rate** across 133 pages of tested literature.
- Designed a **native ChromaDB integration**, bypassing heavier LangChain abstractions, with strict prompt-level context constraints so the system explicitly declines to answer rather than fabricate when retrieval context is insufficient.

EDUCATION

Prince Shri Venkateshwara Padmavathy Engineering College

Expected 2027

B.E. in Electronics and Communication Engineering (ECE)

CGPA - 7.9/10

Sacred Heart Matriculation Higher Secondary School

2023

Higher Secondary Education (Class XII)

Percentage - 76%

CERTIFICATIONS & ACHIEVEMENTS

- Microsoft Certified:** Azure Fundamentals (AZ-900)
- Salesforce:** Agentforce Specialist
- NPTEL:** Programming in Java (**ELITE Badge**)
- Star Rotaractor** (2024-25) & "Impactful Igniter" – Best Secretary Award (2025-26); **Google Crowdsource** Volunteer

LEADERSHIP & EXTRACURRICULARS

Secretary | Rotaract Club of PSVPEC

2025 – 2026

- Orchestrated community events for **60+ members**, boosting club visibility by **40%** via strategic branding while honing skills in project management and team leadership.

LANGUAGES

Tamil (Native) • **English** (Professional) • **Hindi** (Basics) **Telugu** (Basics) • **Japanese** (Beginner)